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RESEARCH ARTICLE

Prediction of Imbalance Prices Through Gradient Boosting Algorithms: An Application to the Greek Balancing Market

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ABSTRACT An accurate prediction of imbalance prices is crucial for making well-informed decisions within short-term energy markets. This study proposes a two-stage probabilistic framework for the prediction of system imbalance and imbalance price. In the first stage, the quantiles of system imbalance are predicted employing the Quantile Regression Forest algorithm. In the second stage, the quantiles are leveraged to predict the imbalance prices, given the observed correlation between the system imbalance and the imbalance prices. The corresponding imbalance prices for each 15-minute interval are then predicted utilizing a novel methodology that combines Gradient Boosting with Markov switching models. The proposed methodology was evaluated using real data from the Greek balancing market. Comparative analysis with alternative models including a persistence model, Quantile Linear Regression, Long Short-Term Memory, and Quantile Regression Forest reveals that the proposed approach consistently outperforms the rest in terms of forecast accuracy. Specifically, the results demonstrate that the proposed approach outperforms all alternative methods, achieving a 9% improvement in nMAE and a 10.8% reduction in CRPS with respect to the second best method (Quantile Regression Forest). The findings suggest that this approach can significantly improve the decision-making process for market participants, helping mitigate the financial risks associated with imbalance prices.

INDEX TERMS Electricity market, balancing market, machine learning, probabilistic forecasting, power system.

I. INTRODUCTION

A. CONTEXT AND LITERATURE REVIEW

The massive integration of Renewable Energy Sources (RES) in the electricity grid has created an increased level of volatility and unpredictability in the electricity markets. This shift in the energy mix greatly affects the operation of markets close to the energy delivery time and predominantly the balancing market. The balancing market, which is

normally operated by the Transmission System Operator (TSO), plays a critical role in managing real-time supply and demand to ensure grid stability and economic efficiency [1]. The difference between supply and demand is defined as the imbalance of the system. ENTSO-E has identified the stochasticity of RES and demand close to real time as one major source of increasing system imbalances in European networks [2]. Other sources that have been identified as increasing the system imbalance are attributed to system factors (e.g. outage of a major production unit, loss of a transmission line) or market factors (e.g. inefficient market

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design). Thus, the increase in imbalance volumes leads to increased activation of reserves in the system, driving up imbalance prices [3].

The problem of system imbalance forecasting has not been studied extensively, but has gained increasing attention during the last few years. In [4], a statistical approach is presented for predicting system imbalance. The developed models cannot accurately capture the trend, due to the non-stationarity of the data. In [5] and [6], Machine Learning (ML) models are employed for system imbalance prediction in the Czech and Spanish power systems, respectively. The key explanatory variables are also identified and the benefits of integrating ensemble techniques for prediction are highlighted. In [7], an intrahour imbalance forecasting tool based on ensemble learning techniques is developed for the Norwegian TSO, which provides significant improvements. Finally in [8], the authors have developed a deterministic system imbalance tool, for multiple forecasting horizons. Different algorithms are compared and applied to the Greek power system. Ensemble techniques like Random Forest (RF), also utilized in this study, demonstrate improved forecasting abilities in the field, confirming findings from the existing literature.

In terms of electricity prices, European markets follow a zonal pricing paradigm. Each zone typically represents a distinct area with its own supply-demand dynamics and network constraints. Electricity prices can vary between these zones due to factors such as generation capacity, demand patterns and transmission constraints [9]. Focusing on the balancing market, which is not yet coupled throughout Europe and is operated by each national TSO independently, single- or dual-pricing mechanisms are employed. In the single-pricing scheme, all participants receive the same price for their balancing actions. Thus, the Imbalance Price (IP) is provided as a remuneration to the Balancing Service Providers (BSPs) for the activation of the reserves and as a penalty (or reward) to the deviating Balancing Responsible Parties (BRPs). In the cases where the deviation of BRPs affects positively the system imbalance, the participant receives the IP as a reward while in the cases where the system imbalance is negatively affected the participant has to pay the IP as a penalty measure. Since BRPs are incentivized to perform self-regulation, the single-pricing scheme has been preferred over dual pricing over the last few years [10].

Historically, Electricity Price Forecasting (EPF) has predominantly relied on deterministic methods, yet these often overlook the inherent uncertainty within the market [11]. This challenge is particularly pronounced in the balancing market, where price fluctuations are greatly influenced by sudden shifts in supply and demand dynamics.

Accurate prediction of imbalance prices is crucial in improving the bidding decisions of market participants (including RES producers, conventional producers, electricity suppliers, energy storage operators), since imbalance prices affect i) the penalty paid or remuneration received by

the market participants in cases where they constitute BRPs, and ii) the remuneration received by the market participants in cases where they constitute BSPs. Such an accurate prediction of imbalance prices becomes more challenging as the integration of RES increases, since their inherent variability and limited predictability increase the volume and frequency of system imbalances.

Probabilistic forecasting offers a spectrum of potential outcomes along with their associated probabilities, providing a more holistic understanding of future uncertainty. Despite its advantages, there have been limited studies of probabilistic forecasting for imbalance prices in the literature, with existing studies mainly focusing on more stable and predictable markets. In [12], the authors emphasize the importance of forecasting the prices of imbalances using various neural network architectures to manage the volatility introduced by renewable energy sources. In [13], a very short-term probabilistic forecasting approach is introduced for imbalance prices, using advanced statistical methods crucial for decision making close to delivery time. The authors in [14] present a novel two-step probabilistic approach tailored to forecast future imbalance prices in the Belgian market, showcasing its effectiveness over other methods. In [15], a hidden Markov model (HMM) was proposed to predict prices and trading volumes, emphasizing its superior predictive capabilities for short- and medium-term forecasts. The impact of forecasting errors in wind and solar energy on market imbalances and spot prices is explored in [16], advocating for improvements in forecast accuracy and transparency. In the context of transformer-based models, the authors in [17] develop a transformer-based model for real-time price prediction that also assesses the importance of input features, which helps short-term trading strategies. Focusing on the factors that affect real-time prices, the authors in [18] analyze the non-linear dynamics affecting system imbalance prices in the Great Britain electricity market, highlighting the advantages of a regime-switching model over traditional models. In the same context, authors in [19] analyze the dynamics and characteristics of the imbalance sign in the Italian electricity market, developing models to predict these dynamics effectively. In view of the trading strategy, in [20] a novel automated trading strategy is introduced using a mixture model to predict system imbalance prices, validated by real-time market data from the Belgian energy market. The authors in [21] discuss the performance of a Long Short-Term Memory (LSTM) network model to forecast prices and volumes in the Nordic balancing market, noting its competent forecasts but limited added value. To construct scenario trees in stochastic optimization, authors in [22] use SARIMA and Markov processes to model real-time balancing power market prices, especially for the generation of price series. Lastly, in [23] an evaluation of various forecasting models for balancing market prices is presented, indicating that none provided effective day-ahead point forecasts, which underscores the challenges in predicting market prices with current models.

B. PAPER CONTRIBUTIONS

Although the existing literature acknowledges the efficacy of tree-based gradient boosting algorithms and Markov regime models individually for forecasting balancing prices in energy markets, a notable gap persists in their combined application. Despite their demonstrated suitability for capturing the volatility of price dynamics in the balancing market, no previous study has integrated these powerful techniques into a relevant research study.

This paper aims to fill the literature gap by introducing a novel probabilistic forecasting framework aimed at the Greek balancing market. To the best of our knowledge, this study represents the first attempt to forecast imbalance prices within a central dispatch single price balancing market model. The approach leverages gradient-boosting algorithms, including regime-switching models to model volatility in the prices. The contributions of the paper can be summarized as follows:

- A two-stage forecaster that produces probabilistic predictions both for the system imbalance and the imbalance price. Incorporating system imbalance as a predictor in the imbalance price forecasting problem enhances the predictive power of our model, marking a novel contribution to the existing literature.
- Combination of a gradient boosting algorithm with a Markov switching model for the prediction of imbalance price. Although these techniques have proven effective in capturing the price dynamics in the balancing market, no prior research has integrated them into a comprehensive study.
- Comprehensive comparison with state-of-the-art methods in predicting the imbalance price. Specifically, state-of-the-art methods include time series techniques, tree-based algorithms, and neural networks. The proposed forecasting framework achieves the highest accuracy in both probabilistic and deterministic predictions.
- This is the first study dedicated to forecasting imbalance prices within the Greek balancing market. Given that the current structure of the Greek electricity market has been recently established, our work provides a foundational framework for understanding and predicting imbalance price dynamics in this market.

C. PAPER ORGANIZATION

The subsequent sections of this paper are organized as follows. The fundamental principles of the Greek balancing market along with the problem setting and the proposed forecasting framework are described in Section II. The analysis dataset and the accuracy metrics are presented in Section III. Quantitative results and relevant discussions are provided in Section IV. Finally, Section V summarizes the conclusions of the paper.

II. METHODOLOGY

In this section, the operating principles of the Greek balancing market are first introduced. Then, we present the

methodology of the forecasting framework along with a mathematical formulation of the models employed.

A. THE GREEK BALANCING MARKET

The Greek energy market has been in compliance with the relevant European directive, known as the Target model, since November 2020. According to Law 4512/2018 the wholesale electricity markets are defined as (a) Forward Capacity Market (FM), (b) Day-Ahead Market (DAM), (c) Intraday Market (IDM), and (d) Balancing Market (BM) [24]. In November 2022, Greece entered Europe's cross-border intraday continuous market (XBID), through a coupling with Italy and Bulgaria [25]. The launch of Continuous Intraday Market trading is a milestone for Greece's RES sector as a large percentage of green power plants will, as a result, immediately fully assume balancing market responsibilities since they can trade up to 1 hour prior to the delivery. Continuous trading will accompany the existing supplementary regional Intra-Day Auctions (IDAs), which serve as the successor of Complementary Intraday Auctions (CRIDAs) [26].

The Hellenic Energy Exchange (HEnEx) is the operator of the DAM and IDM market, while the operation of the BM has been assigned to the Independent Power Transmission Operator (IPTO). This differentiation adheres to the European standards, as the market solutions provided by HEnEx do not include the network constraints limitations, which are integrated into IPTO's solution for the BM.

The purpose of BM is to maintain the real-time balance between production and demand, while ensuring the uninterrupted supply of energy and considering the commitment to the schedule from the previous markets. The operating principle of the Greek BM includes central dispatch of the operating units, executing a security-constrained unit commitment model [27]. In general, the central dispatch model is adopted in a small number of European countries (e.g. Greece, Hungary, Ireland, Poland); in such cases, the impact of locational market imbalances can pose a threat to the system's security. Furthermore, since 2021, IPTO has participated in the European International Grid Control Cooperation (IGCC) for the imbalance netting process and is actively involved in implementation projects for establishing the European platforms MARI and PICASSO, which facilitate the exchange of balancing energy from manual and automatic frequency restoration reserves [28]. Three different sub-markets are defined within the BM, namely Balancing Capacity Market, Balancing Energy Market and Market Settlement. The overview of different timeframes of the wholesale and balancing markets is presented in Figure 1.

In this study, we focus on the Balancing Energy market in which prices and quantities, for balancing supply and demand, are determined. This market includes the activation signals on both automatic and manual frequency restoration reserves, known as aFRR and mFRR respectively. The aFRR dispatch orders are granted every 4 seconds using the

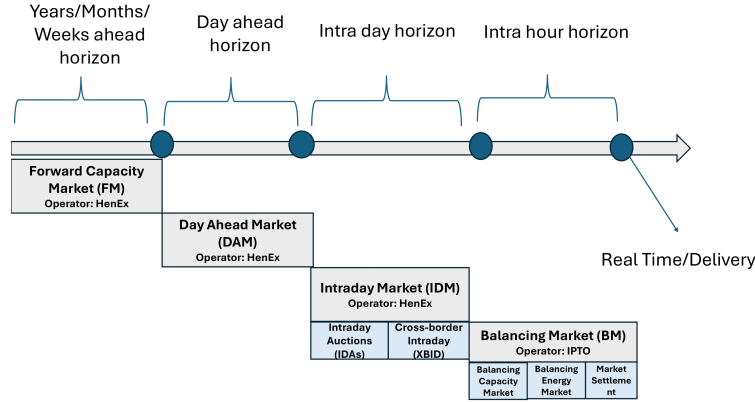


FIGURE 1. Overview of different timeframes of the wholesale and balancing markets in Greece.

Automatic Generation Control (AGC), while for the mFRR the dispatch orders are given every 15 minutes following IPTO's estimation on the system conditions [29].

B. PROBLEM SETTING: SYSTEM IMBALANCE AND IMBALANCE PRICES

System imbalance refers to the net difference between the total activated upward and downward energy. Mathematically, it can be expressed as:

$$SI = ABE_{up} - ABE_{down} \quad (1)$$

where SI denotes the system imbalance, ABE_{up} represents the total activated upward balancing energy, and ABE_{down} corresponds to the total activated downward balancing energy.

A positive system imbalance ($SI > 0$) indicates a shortfall in power supply, leading to a drop in system frequency. To mitigate this, the TSO may activate additional upward reserves, enable demand-side response mechanisms, or import power from neighboring systems. Conversely, a negative system imbalance ($SI < 0$) signifies an excess of generated power over demand, leading to a rise in system frequency. In such cases, the TSO deploys downward reserves, or exports surplus energy to other regions. The effective management of system imbalance is crucial for enhancing the resilience and economic efficiency of power system operations, particularly in the context of increasing RES penetration.

To ensure economic efficiency and grid stability, an imbalance settlement mechanism is employed to financially incentivize BRPs to minimize deviations from their scheduled energy production or consumption. The imbalance settlement period, t , has been set to 15 minutes, aligning with standard market and operational practices [30]. For the calculation of the imbalance price, $IP(t)$, the weighted average price of the activated balancing energy in the predominant direction (both for aFRR and mFRR) is considered. The formula for the

imbalance price is given in Equation 1 [29].

$$IP(t) = \frac{\sum_{BSP_s} ABEC(t)^{mFRR_{pred_dir}} + \sum_{BSP_s} ABEC(t)^{aFRR_{pred_dir}}}{\sum_{BSP_s} ABE(t)^{mFRR_{pred_dir}} + \sum_{BSP_s} ABE(t)^{aFRR_{pred_dir}}} \quad (2)$$

where $ABEC(t)^{mFRR_{pred_dir}}$ and $ABEC(t)^{aFRR_{pred_dir}}$ represents the mFRR and aFRR debit respectively in Euros while $ABE(t)^{mFRR_{pred_dir}}$ and $ABE(t)^{aFRR_{pred_dir}}$ represents the mFRR and aFRR activated balancing energy in MWh. All the terms in Equation 1 refer to the predominant direction of the system. In general, the imbalance price is expected to reflect system conditions and provide the appropriate economic signals to market participants. When the system is short (i.e., excess consumption) the imbalance price is typically higher than the respective DAM price, as the cost of procuring additional balancing energy increases due to the need for rapid adjustments. In contrast, when the system is long (i.e., excess generation) the imbalance price is expected to be lower than the DAM price, as surplus energy must be managed, often leading to lower prices in real-time balancing markets.

C. PROPOSED FORECASTING FRAMEWORK

The proposed forecasting framework aims at generating probabilistic forecasts of the two quantities elaborated in Section II-B, namely system imbalance and imbalance price, for each Imbalance Settlement Period (ISP). The overall architecture of the proposed framework is presented in Figure 2 and contains 2 distinct stages / forecasters, along with the corresponding categories of explanatory variables (inputs). The upper (Stage A) forecaster predicts the system imbalance, while the lower (Stage B) forecaster predicts the imbalance price. The two forecasters are interconnected through the output of the Stage A forecaster, namely the quantiles of system imbalance, which constitutes an input for the Stage B forecaster.

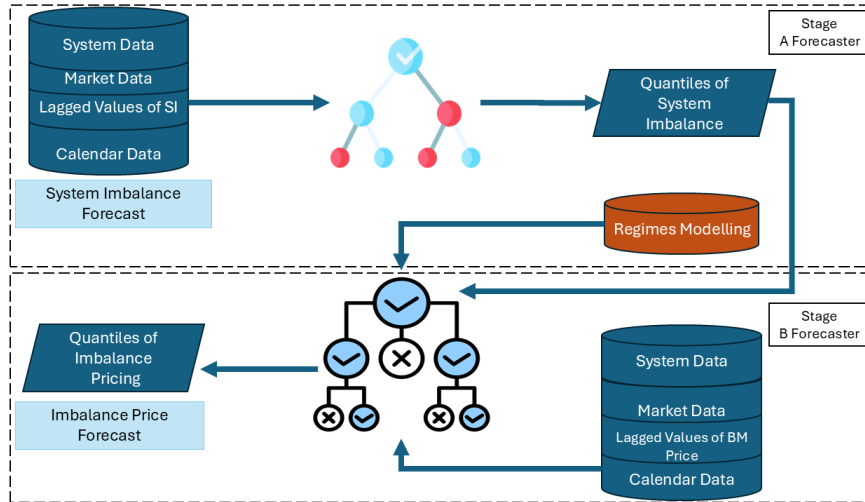


FIGURE 2. Architectural framework of the 2-stage forecasting process; system imbalance and imbalance price forecaster.

For the probabilistic prediction of the imbalance of the system, an extension of the model presented in [8] is adopted. The proposed algorithm is the Quantile Regression Forest (QRF). QRF is a robust method against system imbalance outliers, with a short training time compared to deep learning approaches, which usually also require larger datasets. The explanatory variables in the forecasting framework are divided into regression and categorical types. The regression variables focus on key system and market data, while categorical variables include information such as day type (e.g., working day) and hour index (e.g., morning hours) to address specific seasonal patterns that emerge. The detailed list of inputs for stage A Forecaster is given below:

- **Market data:** This includes price and volume information from different electricity markets (Day-Ahead Market, Intraday Market, and Cross-Border Intraday Continuous Market).
- **System data:** This consists of electricity demand (load) forecasts and renewable energy generation (production) forecasts.
- **Lags of system imbalance:** Past imbalances in the power system are used as inputs to predict future imbalances.
- **Calendar data:** Time-related factors such as the hour of the day, quarter of the year, and type of day (e.g., weekday, weekend, holiday) impact electricity consumption and generation.

For the probabilistic prediction of imbalance price, a novel approach of combining the extreme gradient boosting (XGBoost) algorithm along with Markov switching model is utilized. The two regimes of the BM price are dictated by the net system imbalance, which is determined by the first sub-forecaster. In general, when the system is short (and there is a need for upward regulation), the BM price is usually higher compared to the DAM price. On the other hand, when the system is long (need of downward regulation) the BM price is

usually lower. We employ this information to further enhance the forecasting accuracy of the gradient boosting algorithm along with the system imbalance prediction quantiles of the previous step. Market, system and calendar data are utilized again as predictors for the imbalance price prediction - similar to the stage A Forecaster. Finally, the additional inputs for stage B Forecaster are presented below:

- **Quantiles for system imbalance prediction:** This is the probabilistic output of stage A Forecaster.
- **Lags of imbalance prices:** Past imbalance prices are used as inputs to forecast future imbalance prices.
- **Regime state analysis:** Identifies different market conditions (e.g., high-price or low-price states) based on system imbalance, enhancing the forecasting model.

D. GRADIENT BOOSTING AND MARKOV-SWITCHING MODELS

Extreme Gradient Boosting (XGBoost) is an effective, robust gradient boosting tree-based algorithm, which has proven to be a competent problem solver for applications in time series forecasting. XGBoost algorithm has been employed in many forecasting applications on load [31], RES [32] and price [33] forecasting. Based on the gradient direction of the loss function, it generates a weak learner at each step and accumulates it throughout the entire model. The trees are then built additively and not independently as happens in other algorithms like Random Forest (RF). The objective function is normalized to prevent overfitting:

$$\text{Objective}(\phi) = \sum_{i=1}^n \text{loss}(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (3)$$

where $\text{loss}(y_i, \hat{y}_i)$ is the loss function measuring the difference between the actual value y_i and the predicted value $\hat{y}_i$, $\Omega(f_k)$ is the regularization term for the k -th tree and f_k represents the k -th tree.

The detailed formulation of the loss function is also given:

$$\text{loss}(y_i, \hat{y}_i) = \sum_{i=1}^n L(y_i, \hat{y}_i) + \sum_{k=1}^K \gamma \cdot T + \frac{1}{2} \lambda \sum_{k=1}^K ||w_k||^2 \quad (4)$$

where $L(y_i, \hat{y}_i)$ is the individual loss function for sample i , γ and λ are regularization parameters, T is the number of leaves in the tree, and w_k represents the weights associated with the leaf nodes in the k -th tree.

In our case, the L_2 loss function was selected, while the set of the necessary hyperparameters of the algorithm is tuned using cross-validation [34]. The parameters are given in Section IV.

In the context of a Markov switching model, the system is assumed to transition between different states or regimes over time, and each regime may have its own set of parameters governing the behavior of the system [35]. Markov switching models are then appropriate when input data is prone to flipping between two or more states.

The mathematical formulation of Markov-switching model has the following form:

$$y_t = c_s + x_t \alpha_s + z_t \beta_s + \varepsilon_s \quad (5)$$

where y_t is the dependent variable which describes the dynamic behaviour of the examined variable (BM price) at time t ; c_s is the intercept for regime $s = \{1, \dots, S\}$; x_t is the vector of exogenous invariant variables at time t ; α_s are the invariant coefficients for regime $s = \{1, \dots, S\}$; z_t is the vector of exogenous dependent variables for regime $s = \{1, \dots, S\}$; β_s are the dependent coefficients for regime $s = \{1, \dots, S\}$; and ε_s is an Independent and Identically Distributed (I.I.D assumption) normal error term, capturing the random nature of generation and demand dynamics for regime $s = \{1, \dots, S\}$. In the proposed model, we test the determinism of balancing price in the market using the variables depicted in Figure 2.

By incorporating the Markov switching model, XGBoost can adaptively produce predictions based on the current state of the system. This integration allows XGBoost to capture dynamic dependencies and temporal patterns which influence the imbalance price, leading to more accurate and robust predictions.

The temporal dependencies in the electricity balancing market arise from both system operations and regulatory influences, including market rules and policies. Variations in supply and demand dynamics across different market intervals can cause the system to transition between long and short state multiple times within a single day. With the increasing penetration of RES, the stochastic nature of weather conditions, and shifts in demand patterns, these transitions are expected to become more pronounced in the coming years. Markov switching models can effectively identify these regimes by enriching the feature space, providing XGBoost with additional context to make informed predictions.

XGBoost is good at learning non-linear relationships and interactions while Markov switching models identify state-dependent patterns as well as regime changes; therefore, together they complement each other in an apposite way. This integration is particularly useful in imbalance price forecasting as state of the system is very dynamic and the relationships are often too complicated for standard modeling techniques to capture. In effect, the combined approach allows the creation of predictive systems that can cope with changed circumstances in the market and which explain the cause for the imbalance price variability, improving the understanding of the processes in the electricity markets.

III. CASE STUDY

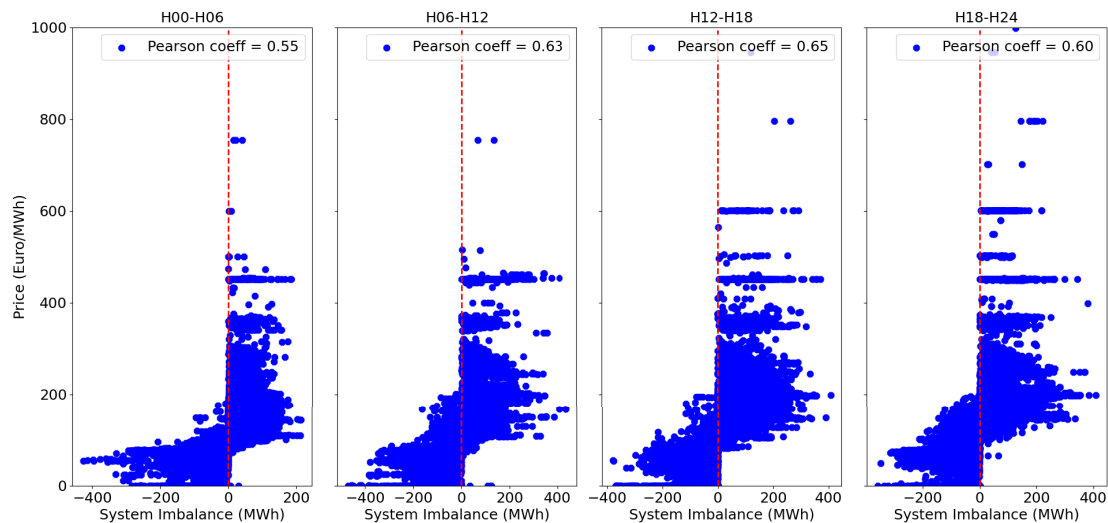
A. DATASET OVERVIEW

The dataset consists of 15-minute resolution data from 1-1-2023 to 1-1-2024. Day-ahead and intraday market data (prices and volumes) are obtained from HEnEx [36], system data from ENTSO-E Transparency Platform [37] and balancing market data (prices and volumes of imbalances) from IPTO [38]. Data before 2023 are disregarded because of XBID's launch in November 2022. Utilizing this data would likely distort the forecasting accuracy because it reflects a period before the implementation of XBID, during which participants may have employed different strategies, potentially introducing biases or inconsistencies in the predictive model. In Table 1, the key statistics of the dataset are presented. These statistics indicate an overall satisfactory performance of the electricity market. Specifically, the DAM operates smoothly and has sufficient liquidity while the IDM includes trivial volumes and limited liquidity. The continuous intraday market (XBID) presents higher volumes, since market participants can react to the latest available information close to real-time. Concerning BM, a sufficient trade volume is observed that is expected to increase due to the continuous integration of RES into the power system, creating an even more volatile price environment for the imbalance price within the market [39].

In Figure 3, we notice the correlations between system imbalance and imbalance price for different hours of the day, divided into four time blocks. The Pearson correlation coefficients are 0.63 for the morning block (H06-H12) and 0.65 for the afternoon block (H12-H18). The trend is similar across all subfigures, indicating that the utilization of system imbalance as an explanatory variable can improve the forecasting accuracy of the algorithm [40]. It should be noted that high imbalance prices can be observed even in hours with relatively low volumes of total system imbalance. The total system imbalance can be small, but the high activation volumes of upward and downward reserves can lead to increased imbalance prices. For example, on 29.6.2023 at 19.30 the imbalance price accounts for 451 €/MWh, while the system imbalance is 28.43 MWh. However, the activation of the upward and downward reserves for this period represented 122.75 and 94.89 MWh, respectively. This emphasizes that

TABLE 1. Key statistics Greek energy market 2023.

Parameter	Average	Standard Deviation	Range
DAM price (€/MWh)	119.7	50.4	384.8
DAM volume (MWh)	6015.4	1327.6	7603.1
ID price (€/MWh)	112.2	48.6	795.1
ID volume (MWh)	164.4	93.9	930.8
XBID price (€/MWh)	93.9	23.5	311.7
XBID volume (MWh)	452.7	302.5	9150.4
BM price (€/MWh)	124.9	102.7	1702.3
BM regulation Up (MWh)	75.3	63.7	452.7
BM regulation Down (MWh)	75.2	62.7	449.5
System Load (MWh)	4463.7	1266.9	8454.3
RES production (MWh)	1189.3	756.1	3521.3

**FIGURE 3.** Correlation of system imbalance with imbalance price.

even when a control area experiences a positive system imbalance, there will be a requirement for activating both upward and downward reserves, primarily due to network constraints.

In Figure 4, the Auto Correlation (ACF) and Partial Auto Correlation (PACF) of the Greek imbalance price are shown. The ACF indicates the correlation of the imbalance price at time t with the corresponding values at time $t-k$ ($k = 15$ minutes). The PACF measures the correlation between the imbalance price and the indicated lag without accounting for the effect of the intermediate lags. The red dashed lines represent the significance threshold at the 5% level. Lags that fall within this range indicate that the corresponding autocorrelations are not statistically significant, suggesting that these specific lags are not important in explaining the variation in the data.

We observe from the PACF plot that the first 4 lags present a high correlation with the imbalance price. However, the first 3 lags are not known at the forecast issue time, since the forecast horizon of the study is 1 hour. Thus, only the 4th

lag can be used as an explanatory variable in the forecasting framework.

We also perform the Augmented Dickey-Fuller (ADF) test, both for the system imbalance and imbalance price, to examine the stationarity of the predicted variables. The test is applied to detect the presence of a unit root in both series and remove any seasonal components or trends if detected. In both cases, the null hypothesis of a unit root was rejected, indicating that the predicted time series are stationary. Notably, the ADF test statistic for both series was less than the 5% critical value. Specifically, the ADF test statistic for system imbalance was -12.69 , well below the 5% critical value of -2.86 . Similarly, the ADF test statistic for the imbalance price was -16.69 , compared to the 5% critical value of -2.53 .

We also present in Figure 5, the yearly, daily and hourly seasonality of the system imbalance along with the corresponding imbalance prices. The blue bars indicate the total system imbalance, while the black line shows the average imbalance price. There are considerable seasonalities

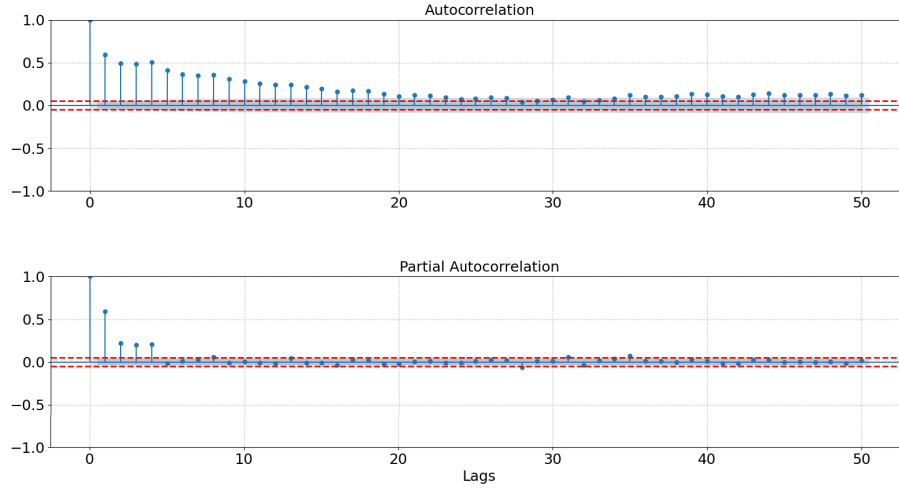


FIGURE 4. ACF and PACF of the BM price.

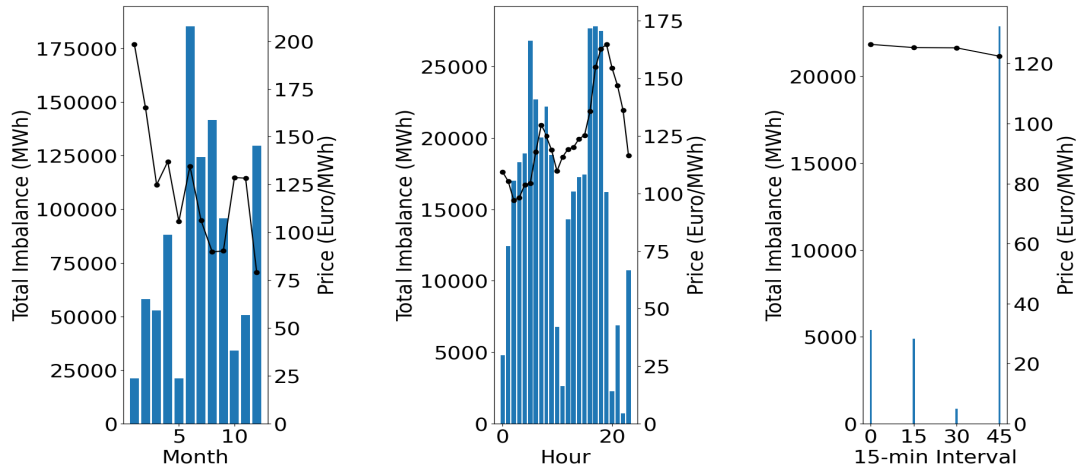


FIGURE 5. Seasonalities of system imbalance and imbalance price Left: Yearly Middle: Daily Right: Hourly.

within both the yearly and daily cycles. The imbalance prices have two peaks; a morning peak around 9 am and an evening peak in the 6 pm to 8 pm range. The volume of system imbalance also follows the same trends for similar hours. In terms of yearly seasonality, it is clear that during the summer months, the highest volumes of system imbalance are observed. This can be attributed to the increased load of the system and the penetration of RES into the system. In particular, RES connected to the LV grid (installation capacity of 7GW at the end of 2023 [41]) contribute mainly to the increased volumes of system imbalance during the summer months. Finally, the highest system imbalance is observed in the last quarter of the hourly cycle, as plants have to adjust their generation levels to comply with the market dispatch of the following hour.

B. FORECASTING ACCURACY METRICS

In probabilistic forecasting, reliability and sharpness are the two fundamental concepts for the evaluation of a forecast.

Reliability refers to the consistency of the forecast value with the actual ones, while sharpness gives information on concentrating the forecast probabilities close to the actual value.

In this study, we employ five accuracy metrics; two probabilistic and three deterministic. The probabilistic metrics are the Continuous Ranked Probability Score (CRPS) and the Pinball Loss Function (PLF) [42] while the deterministic metrics are Root Mean Square Error (RMSE), Mean Absolute Error (MAE) and the normalized Mean Absolute Error (nMAE) [43], [44], [45].

Unlike the PLF, the CPRS does not focus on any specific point of the probability distribution but considers the distribution of the forecasts as a whole. It measures both reliability and sharpness and is expressed in the same unit as the observed variable. For a random variable X , the formulation is given:

$$\text{CRPS}(F, x) = \int_{-\infty}^{\infty} [(F(y) - \mathcal{H} \cdot (y - x))^2] dy \quad (6)$$

where x is the observed value of imbalance price (p_{imb}), F is the cumulative distribution function, $F(y) = P[X \leq y]$ and $\mathcal{H}$ represents the Heaviside step function and takes the value of 1 if $y \geq x$ and 0 otherwise.

PLF is an asymmetric performance metric and penalizes deviations from the quantile differently depending on whether the observed result y is above or below the forecasted quantile q . Like CRPS, a lower pinball loss indicates a better forecast accuracy.

$$\text{Pinball Loss}(q, y) = \begin{cases} (y - q) \cdot \tau, & \text{if } y \geq q \\ (q - y) \cdot (1 - \tau), & \text{if } y < q \end{cases} \quad (7)$$

where q represents the forecasted quantile, y is the observed value of the imbalance price (p_{imb}) and τ is the quantile level, which ranges between 0 and 1.

Concerning deterministic metrics, we firstly employ Root Mean Squared Error (RMSE), a widely used metric that quantifies the standard deviation of prediction errors. By squaring the differences, RMSE penalizes larger errors more heavily, making it particularly useful for highlighting significant deviations from actual values.

Additionally, we use Mean Absolute Error (MAE), which measures the average magnitude of prediction errors while treating all deviations equally. Unlike RMSE, MAE does not disproportionately emphasize larger errors, providing a balanced measure of overall accuracy.

Finally, we incorporate Normalized Mean Absolute Error (nMAE), which scales MAE by the average value of the dataset. Expressed as a percentage (%), nMAE is applied specifically to the 50th quantile (median value), allowing for meaningful comparisons across datasets of different scales.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (8)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (9)$$

$$nMAE = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{\bar{y}} \quad (10)$$

where n represents the number of observations, y_i is the observed value of imbalance price (p_{imb}), $\hat{y}_i$ is the corresponding forecasted value and $\bar{y}$ is the mean of the observed values.

C. EXAMINED STATE-OF-THE-ART TECHNIQUES

Finally, the proposed price forecast model is compared against four state-of-the-art techniques to evaluate its robustness and accuracy:

- Persistence [46]: The average value of the last 4 imbalance prices for the same hour index is taken, while the variance is determined by the Gaussian distribution.
- Quantile Regression [47], [48]: This is an extension of regular linear regression that estimates the conditional median of the target value.

- Long Short-Term Memory (LSTM) [49] networks are a type of Recurrent Neural Network (RNN) [50] designed to better capture long-term dependencies in sequential data. They are particularly effective in addressing the vanishing and exploding gradient problems often encountered in traditional RNNs.
- Quantile Regression Forest [51]: This is an ensemble learning method that combines decision trees to estimate conditional quantiles. The decision trees are constructed independently, and each tree's predictions are aggregated to provide reliable estimates of conditional quantiles.

IV. RESULTS AND DISCUSSION

In this section, we present the results of the forecasting framework for system imbalance and imbalance price forecasting. We conduct the study on an Intel CoreTM i7-3555 CPU @1.7 GHz with 16 GB RAM. The forecasting models used are implemented in Python 3.6 using the Scikit-learn and TensorFlow libraries. The data spans from 1-1-2023 to 1-1-2024 with a resolution of 15 minutes, resulting in a sample size of 35040 rows. The dataset is divided using a ratio of 70-20-10 for the training, validation and testing set, respectively. A forecasting horizon of 1 hour is selected, which corresponds to 4 time steps ahead, since all examined market and system predictors (Section II-C) are accessible up to a minimum of 1 hour before the delivery time.

In order to optimally tune the hyperparameters of the proposed model, we employ Grid Search, an exhaustive search method that evaluates multiple combinations of hyperparameters within predefined ranges. Specifically, this search is performed using 5-fold cross-validation on the training set, and the hyperparameters' combination that yields the best performance -based on the RMSE and MAE metrics- is selected. The hyperparameters' search ranges, along with their optimally tuned values (indicated in bold font) are presented below:

Quantile Regression Forest (QRF) – System Imbalance Forecaster:

- Number of Trees: {100, 200, **500**, 1000}
- Maximum Depth: {5, 10, **15**, 20}
- Minimum Sample Leaf Size: {1, **3**, 5, 10}
- Maximum Number of Features: {2,4,**5**,7}

XGBoost – Imbalance Price Forecaster:

- Number of Trees: {100, 200, **500**, 1000}
- Learning Rate: {0.01, 0.1, **0.3**}
- Maximum Depth: {6, 8, 10, **12**}
- Gamma: {0, 0.1, **0.2**, 0.3}
- Maximum Number of Features: {2,4,**5**,7}

A. SYSTEM IMBALANCE PREDICTION

The results of system imbalance forecasting are depicted in this section. Table 2 presents the results of the Quantile Regression Forest for the prediction of system imbalance. As also shown in [8], the complex nature of

TABLE 2. Metrics on system imbalance prediction.

Model	Probabilistic						Deterministic		
	Quantile 5%	Quantile 25%	Quantile 50%	Quantile 75%	Quantile 95%	CRPS	RMSE	MAE	nMAE (%)
QRF	5.8	17.81	21.93	17.19	5.25	40.12	32.6	25.2	14.7

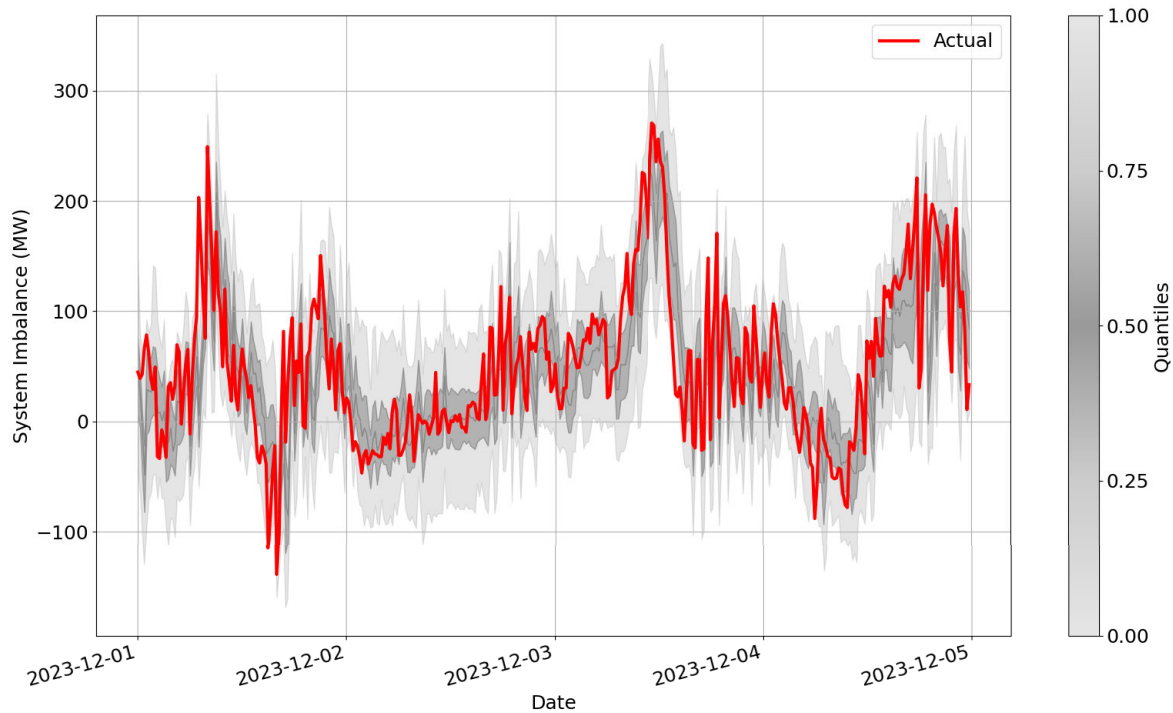


FIGURE 6. Forecasting results for system imbalance under the proposed QRF model.

TABLE 3. BM price forecasting metrics.

Model	Probabilistic						Deterministic		
	Quantile 5%	Quantile 25%	Quantile 50%	Quantile 75%	Quantile 95%	CRPS	RMSE	MAE	nMAE (%)
Baseline Persistence	10.74	31.65	40.50	31.35	12.49	61.47	55.31	36.98	17.4
Quantile Regression	7.21	23.85	29.83	25.01	12.15	42.67	51.82	30.56	12.8
LSTM	7.02	22.95	27.44	24.19	13.01	42.13	50.13	30.77	12.14
QRF	6.02	20.37	25.68	22.49	14.57	41.54	46.22	25.71	11.05
XGBoost+Regimes	4.69	19.71	23.93	20.67	7.87	37.05	39.13	21.16	10.06

the system imbalance forecasting problem and the existing non-linearities can accurately be captured by the ensemble training method. The achieved CRPS indicates that the forecasted quantiles of system imbalance closely match the actual values. The produced forecast, both in terms of reliability and sharpness, demonstrates a high degree of accuracy and precision. The PLF metric shows that there is no great discrepancy between the forecasted and actual values for every quantile. This can also be verified by Figure 6, in which a graph is presented showing the results for the first 5 days of the test set. The light gray colour indicates the 5th and 95th quantile while the darker color indicates the range between the 25th and 75th quantile. The overall trend of the system imbalance is accurately captured by the proposed

algorithm. The spikes in the period shown are correctly identified, with a slight tendency to underestimate when the imbalance is over 250MW. There is a clear cyclic component of the imbalance during the evening hours, as explained in Section III. The stochasticity added to the problem from the low-voltage-connected RES (mainly rooftop solar) can partially explain the decreased forecasting accuracy in these hours.

B. IMBALANCE PRICE PREDICTION

The results of the imbalance price forecasting are presented in this section. Table 3 presents the forecasting results of the compared algorithms for the imbalance price prediction. The baseline persistence model and the quantile regression both

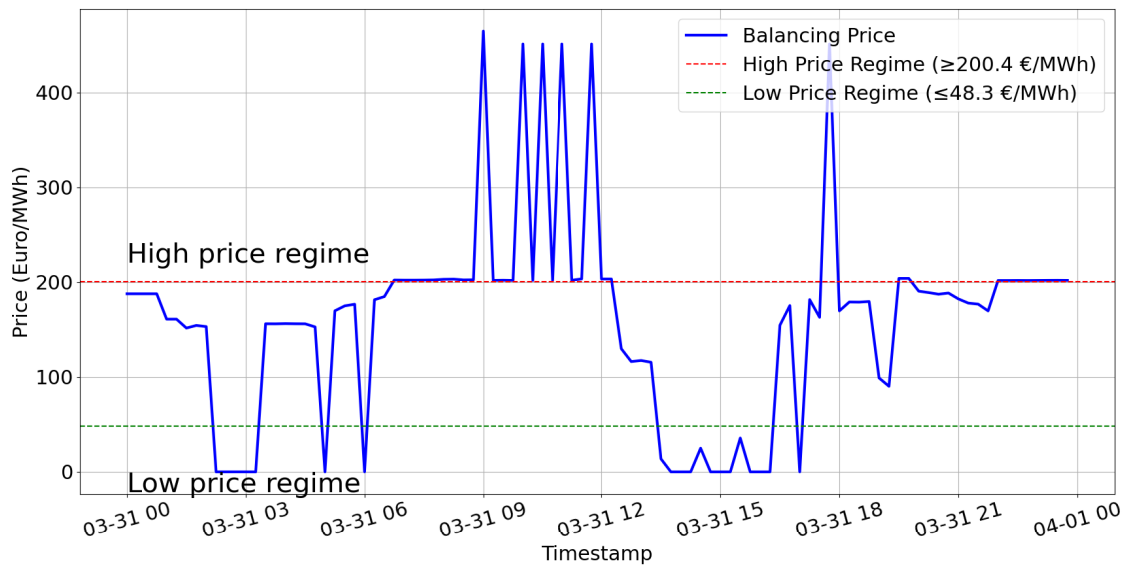


FIGURE 7. Identification of price regimes in the Greek balancing market.

rely on linear relationships between the analyzed predictors and the imbalance price. Thus, they present a greater discrepancy in terms of the forecast accuracy with respect to all the metrics examined. LSTM performs significantly better compared to linear methods, but it appears to be more suitable in capturing long-term dependencies in sequential data. It often struggles with the short-term dependencies and volatile nature of balancing market price fluctuations and fails to adapt quickly to regime changes within short intervals. The QRF algorithm is again deployed here, further improving the forecast accuracy by having greater resilience to the outliers and spikes which are prevailing in the balancing market. However, all benchmarked methods are outperformed by the proposed XGBoost forecasting algorithm. The CRPS metric is improved by 10.8% while the nMAE which measures the accuracy of the 50th quantile is also improved by 8.9% compared to the QRF algorithm. The gradient boosting and regularization techniques employed by the XGBoost algorithm can more accurately forecast price spikes that frequently occur in the balancing market. Markov switching models excel in capturing regime changes inherent in electricity market dynamics, such as shifts in demand patterns, regulatory impacts, and the stochastic nature of renewable energy sources. By incorporating these regime states as additional features, XGBoost gains the ability to adapt its predictions based on the current market conditions identified by the Markov model. This adaptive capability allows XGBoost to refine its predictions more effectively, optimizing forecasting accuracy across different market regimes. This approach provides a more robust framework for imbalance price prediction, demonstrating the value of combining traditional time series models with advanced machine learning techniques.

Specifically, the Markov switching model enhances the performance of the proposed method by providing insights into the duration of each price regime. Figure 7 illustrates the regime-switching behavior of the balancing market price for a specific day. The low price regime indicates that the system is long ($\lambda_{BM} < \lambda_{DAM}$), while the high price regime shows that the system is short ($\lambda_{BM} > \lambda_{DAM}$). The high-price regime ($> 200.4 \text{ €/MWh}$) and low-price regime ($< 48.3 \text{ €/MWh}$) are highlighted using dashed lines to indicate the distinct pricing states.

The low-price regime usually lasts shorter compared to a high-price regime for the Greek power system. Specifically, the duration of a low-price regime lasts for 90 minutes, whereas the high-price regime's duration accounts for 120 minutes. This suggests that once the system enters the high regime, it tends to stay there for a longer period before transitioning to another regime.

Finally, in Figure 8, we present the detailed time series results for the proposed ensemble algorithm. The quantiles are again depicted in the same colors as in Figure 6. In general, the algorithm performs well when there are no extreme price events in the system. It captures very effectively the trends of the fluctuating prices, yet with a small delay particularly due to the inclusion of the imbalance price lag in the forecasting framework. The spikes above 300 €/MWh cannot be captured effectively, since these are usually caused by strategic behavior of market participants. This argument is justified by the findings of [52] and [53], which demonstrate that market participants often engage in strategic bidding by exploiting structural system limitations and sudden faults of generation plants and transmission lines. These strategic behaviors are inherently difficult to predict.

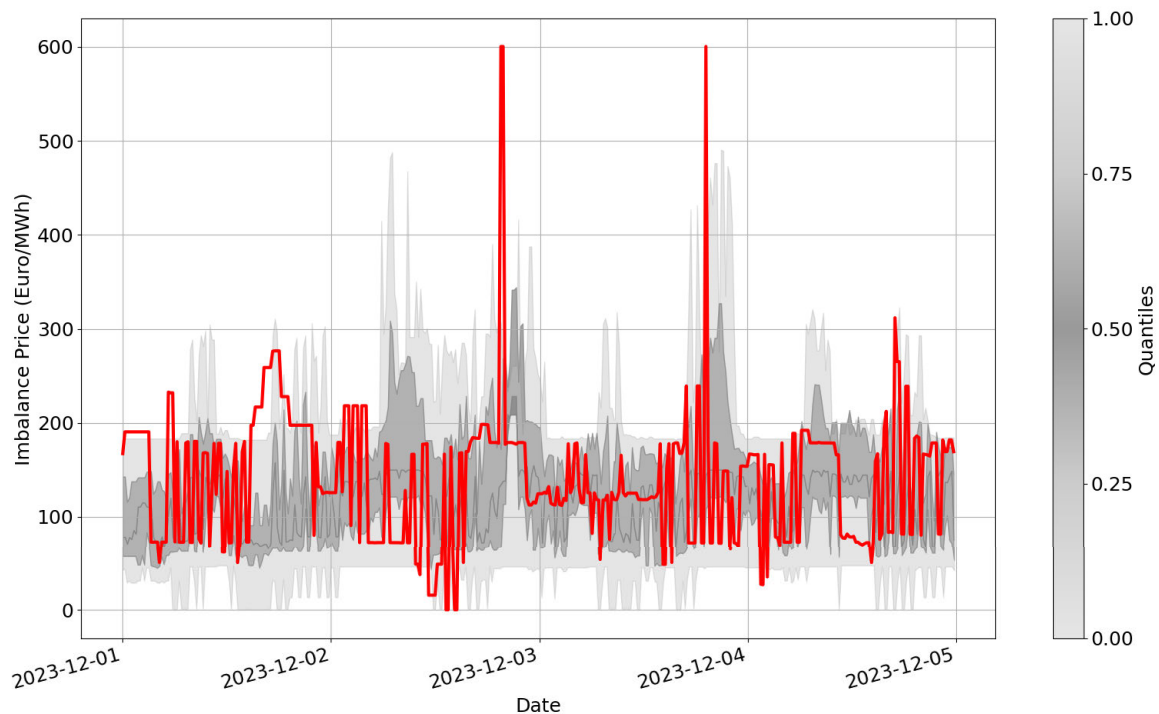


FIGURE 8. Forecasting results for imbalance price under the proposed XGBoost+Regimes model.

V. CONCLUSION

A. KEY LESSONS LEARNED AND IMPLICATIONS TO THEORY AND PRACTICE

In this paper, a complete probabilistic forecasting framework for the prediction of imbalance prices has been presented and analyzed. The forecasting framework is comprised of two sub-forecasters which aim to predict sequentially the imbalance volumes and the imbalance prices. A set of external system and market predictors have been utilized in the analysis. A novel gradient tree-based algorithm, combined with a Markov regime switching model has been proved to provide consistently more accurate forecasts compared to four state-of-the-art methods, both in terms of probabilistic and deterministic prediction of the imbalance prices.

The study highlights the critical role of system imbalance in imbalance price forecasting, reinforcing the close correlation between these two quantities. In practice, improved forecast of imbalance prices can help market participants make more informed bidding decisions, improving risk management and market efficiency.

A key insight from the study is that including system imbalance as a predictor significantly enhances forecasting accuracy. This underscores the importance of integrating real-time system indicators when modeling electricity price fluctuations. Furthermore, traditional forecasting models may struggle to capture the complexity of imbalance prices, necessitating more advanced probabilistic frameworks.

B. LIMITATIONS AND FUTURE RESEARCH DIRECTIONS

An important limitation of the proposed model observed in the case study, is that it cannot effectively capture extreme price events. This is due to the fact that these price spikes are usually caused by strategic behavior of market participants who exploit structural system limitations and sudden faults of generation plants and transmission lines. In addition, the study focuses primarily on net imbalance volume forecasting; predicting upward and downward imbalances separately may further refine the precision of the forecast.

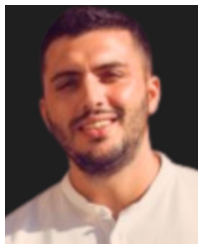
Future research should explore the effects of strategic market behavior and generation / network outages on imbalance prices to further enhance the forecasting accuracy under extreme price events. Specifically, modeling market participants' bids directly, rather than relying solely on imbalance price time series predictions, could help capture strategic behavior more accurately and identify the hours when market participants bid significantly higher than their marginal cost, which may explain some of the extreme price fluctuations.

In addition to this, integrating information from the low voltage grid (e.g., RES production etc.) could substantially improve the RES forecast error, leading to higher precision forecast for the system imbalance. In this context, the prediction of the volumes independently (both upwards and downwards) could also be studied in future work, instead of predicting the net imbalance volume.

REFERENCES

- [1] C. Roumkos, P. N. Biskas, and I. G. Marneris, "Integration of European electricity balancing markets," *Energies*, vol. 15, no. 6, p. 2240, Mar. 2022.
- [2] *Operational Reserve Ad Hoc Team Report*, L. ENTSO-E, Brussels, Belgium, 2012.
- [3] A. Eicke, O. Ruhnau, and L. Hirth, "Electricity balancing as a market equilibrium: An instrument-based estimation of supply and demand for imbalance energy," *Energy Econ.*, vol. 102, Oct. 2021, Art. no. 105455.
- [4] M. P. Garcia and D. S. Kirschen, "Forecasting system imbalance volumes in competitive electricity markets," *IEEE Trans. Power Syst.*, vol. 21, no. 1, pp. 240–248, Feb. 2006.
- [5] Š. Kratochvíl, "System imbalance forecast," Ph.D. dissertation, Dept. Econ., Manage. Humanities, Czech Tech. Univ., Prague, Czechia, 2016.
- [6] C. Contreras, "System imbalance forecasting and short-term bidding strategy to minimize imbalance costs of transacting in the Spanish electricity market," Master's thesis, Dept. Electr. Power Ind., Comillas Pontifical Univ., Madrid, Spain, 2016.
- [7] T. S. Salem, K. Kathuria, H. Ramampiaro, and H. Langseth, "Forecasting intra-hour imbalances in electric power systems," in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, Jul. 2019, pp. 9595–9600.
- [8] K. Plakas, N. Andriopoulos, A. Birbas, I. Moraitis, and A. Papalexopoulos, "A forecasting model for the prediction of system imbalance in the Greek power system," *Eng. Proc.*, vol. 39, no. 1, p. 18, Jun. 2023.
- [9] P. F. Borowski, "Zonal and nodal models of energy market in European union," *Energies*, vol. 13, no. 16, p. 4182, Aug. 2020.
- [10] E. Rivero, J. Barquín, and L. Rouco, "European balancing markets," in *Proc. 8th Int. Conf. Eur. Energy Market (EEM)*, May 2011, pp. 333–338.
- [11] R. Weron, "Electricity price forecasting: A review of the state-of-the-art with a look into the future," *Int. J. Forecasting*, vol. 30, no. 4, pp. 1030–1081, Oct. 2014.
- [12] V. N. Ganesh and D. Bunn, "Forecasting imbalance price densities with statistical methods and neural networks," *IEEE Trans. Energy Markets, Policy Regulation*, vol. 2, no. 1, pp. 1–10, Mar. 2024.
- [13] M. Narajewski, "Probabilistic forecasting of German electricity imbalance prices," *Energies*, vol. 15, no. 14, p. 4976, Jul. 2022.
- [14] J. Dumas, I. Boukas, M. M. de Villena, S. Mathieu, and B. Cornélusse, "Probabilistic forecasting of imbalance prices in the Belgian context," in *Proc. 16th Int. Conf. Eur. Energy Market (EEM)*, Sep. 2019, pp. 1–7.
- [15] I. Dimoulkas, M. Amelin, and M. R. Hesamzadeh, "Forecasting balancing market prices using hidden Markov models," in *Proc. 13th Int. Conf. Eur. Energy Market (EEM)*, Jun. 2016, pp. 1–5.
- [16] S. Goodarzi, H. N. Perera, and D. Bunn, "The impact of renewable energy forecast errors on imbalance volumes and electricity spot prices," *Energy Policy*, vol. 134, Nov. 2019, Art. no. 110827.
- [17] J. Bottieau, Y. Wang, Z. De Grève, F. Vallée, and J.-F. Toubeau, "Interpretable transformer model for capturing regime switching effects of real-time electricity prices," *IEEE Trans. Power Syst.*, vol. 38, no. 3, pp. 2162–2176, May 2023.
- [18] D. W. Bunn, J. N. Inekwe, and D. MacGeehan, "Analysis of the fundamental predictability of prices in the British balancing market," *IEEE Trans. Power Syst.*, vol. 36, no. 2, pp. 1309–1316, Mar. 2021.
- [19] F. Lisi and E. Edoli, "Analyzing and forecasting zonal imbalance signs in the Italian electricity market," *Energy J.*, vol. 39, no. 5, pp. 1–20, Sep. 2018.
- [20] R. Bruneel, M. Schuurmans, and P. Patrinos, "Optimal intraday power trading for single-price balancing markets: An adaptive risk-averse strategy using mixture models," 2024, *arXiv:2402.01215*.
- [21] S. Backe, S. Riemer-Sørensen, D. A. Bordvik, S. Tiwari, and C. A. Andresen, "Predictions of prices and volumes in the Nordic balancing markets for electricity," in *Proc. 19th Int. Conf. Eur. Energy Market (EEM)*, Jun. 2023, pp. 1–6.
- [22] M. Olsson and L. Soder, "Modeling real-time balancing power market prices using combined SARIMA and Markov processes," *IEEE Trans. Power Syst.*, vol. 23, no. 2, pp. 443–450, May 2008.
- [23] G. Klæboe, A. L. Eriksrud, and S.-E. Fleten, "Benchmarking time series based forecasting models for electricity balancing market prices," *Energy Syst.*, vol. 6, no. 1, pp. 43–61, Mar. 2015.
- [24] A. Metaxas, M. Mathioulakis, and M. Lykidi, "Implementation of the target model: Regulatory reforms and obstacles for the regional market coupling," *Eur. Energy Climate J.*, vol. 8, nos. 3–4, pp. 28–43, Jul. 2019.
- [25] *Entsoe Single Intraday Coupling (sidc)*. [Online]. Available: https://www.entsoe.eu/network_codes/cacm/implementation/sidc/
- [26] F. Ioannidis, T. Georgitsioti, K. Kosmidou, C. Zopounidis, and K. Andriopoulos, "A simulation model for imbalance costs of renewable energy aggregators: The case of Greek balancing market," *Energy Econ.*, vol. 142, Feb. 2025, Art. no. 108155.
- [27] J. J. Makrygiorgou, C.-S. Karavas, C. Dikaikos, and I. P. Moraitis, "The electricity market in greece: Current status, identified challenges, and arranged reforms," *Sustainability*, vol. 15, no. 4, p. 3767, Feb. 2023.
- [28] RAE. (2023). *High-level Design of the Mfrr Process in the Greek Balancing Energy Market for Participation in the European Mfrr Platform, Mari, for the Exchange of Balancing Energy From Frequency Restoration Reserves With Manual Activation*. [Online]. Available: <https://www.rae.gr/wp-content/uploads/2023/01/mari-preliminary-design-20221228.pdf>
- [29] *Ipto. Balancing Market Rulebook-ver. 3.0. 2020*, IPTO, Athens, Greece, 2020.
- [30] ENTSOE. (2023). *Entsoe Market Report 2023*. [Online]. Available: <https://eepublicdownloads.entsoe.eu/clean-documents/nc-tasks/ENTSO-EMarketReport2023.pdf>
- [31] Y. Wang, S. Sun, X. Chen, X. Zeng, Y. Kong, J. Chen, Y. Guo, and T. Wang, "Short-term load forecasting of industrial customers based on SVM and XGBoost," *Int. J. Electr. Power Energy Syst.*, vol. 129, Jul. 2021, Art. no. 106830.
- [32] H. Abbasimehr, R. Paki, and A. Bahrini, "A novel XGBoost-based featurization approach to forecast renewable energy consumption with deep learning models," *Sustain. Comput., Informat. Syst.*, vol. 38, Apr. 2023, Art. no. 100863.
- [33] K. Wu, Y. Chai, X. Zhang, and X. Zhao, "Research on power price forecasting based on PSO-XGBoost," *Electronics*, vol. 11, no. 22, p. 3763, Nov. 2022.
- [34] S. Ramraj, N. Uzir, R. Sunil, and S. Banerjee, "Experimenting XGBoost algorithm for prediction and classification of different datasets," *Int. J. Control Theory Appl.*, vol. 9, no. 40, pp. 651–662, 2016.
- [35] S. Kolosok, L. Saher, Y. Kovalenko, and M. Delibasic, "Renewable energy and energy innovations: Examining relationships using Markov switching regression model," *Marketing Manage. Innov.*, vol. 2, no. 1, pp. 151–160, 2022.
- [36] EnEX. *Enex Day-ahead Market Data*. Accessed: Mar. 1, 2024. [Online]. Available: <https://www.enxgroup.gr/el/markets-publications-el-day-ahead-market>
- [37] ENTSOE. *Entsoe Transparency Platform*. Accessed: Mar. 1, 2024. [Online]. Available: <https://transparency.entsoe.eu/>
- [38] IPTO. *Ipto Market Statistics*. Accessed: Mar. 1, 2024. [Online]. Available: <https://www.admie.gr/en/market/market-statistics/detail-data>
- [39] A. Gianfreda, L. Parisio, and M. Pelagatti, "A review of balancing costs in Italy before and after RES introduction," *Renew. Sustain. Energy Rev.*, vol. 91, pp. 549–563, Aug. 2018.
- [40] K. Rawal and A. Ahmad, "Feature selection for electrical demand forecasting and analysis of Pearson coefficient," in *Proc. IEEE 4th Int. Electr. Energy Conf. (CIEEC)*, May 2021, pp. 1–6.
- [41] IEA. (2023). *World Energy Outlook 2023*. [Online]. Available: <https://www.iea.org/reports/world-energy-outlook-2023>
- [42] A. Jordan, F. Krüger, and S. Lerch, "Evaluating probabilistic forecasts with scoringRules," 2017, *arXiv:1709.04743*.
- [43] M. Shcherbakov, A. Brebels, A. Tyukov, T. Janovsky, and V. Anatol, "A survey of forecast error measures," *World Appl. Sci. J.*, vol. 24, no. 24, pp. 171–176, Jan. 2013.
- [44] K. Plakas, I. Karampinis, P. Alefragis, A. Birbas, M. Birbas, and A. Papalexopoulos, "A predictive fuzzy logic model for forecasting electricity day-ahead market prices for scheduling industrial applications," *Energies*, vol. 16, no. 10, p. 4085, May 2023.
- [45] I. K. Bazionis and P. S. Georgilakis, "Review of deterministic and probabilistic wind power forecasting: Models, methods, and future research," *Electricity*, vol. 2, no. 1, pp. 13–47, Jan. 2021.
- [46] S. Dutta, Y. Li, A. Venkataraman, L. M. Costa, T. Jiang, R. Plana, P. Tordjman, F. H. Choo, C. F. Foo, and H. B. Puttgen, "Load and renewable energy forecasting for a microgrid using persistence technique," *Energy Proc.*, vol. 143, pp. 617–622, Dec. 2017.
- [47] R. M. A. Oliveira, Â. M. O. Sant'Anna, and P. H. F. da Silva, "Explainable machine learning models for defects detection in industrial processes," *Comput. Ind. Eng.*, vol. 192, Jun. 2024, Art. no. 110214.
- [48] V. Jensen, F. M. Bianchi, and S. N. Anfinsen, "Ensemble conformalized quantile regression for probabilistic time series forecasting," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 9014–9025, Jul. 2022.

- [49] B. Amiri, A. Haddadi, and K. Farajpour Mojdehi, "A novel hybrid GCN-LSTM algorithm for energy stock price prediction: Leveraging temporal dynamics and inter-stock relationships," *IEEE Access*, vol. 13, pp. 24815–24832, 2025.
- [50] S. Sang, F. Qu, and P. Nie, "Ensembles of gradient boosting recurrent neural network for time series data prediction," *IEEE Access*, early access, May 21, 2021, doi: [10.1109/ACCESS.2021.3082519](https://doi.org/10.1109/ACCESS.2021.3082519).
- [51] F. Houndekindo and T. B. M. J. Ouara, "Prediction of hourly wind speed time series at unsampled locations using machine learning," *Energy*, vol. 299, Jul. 2024, Art. no. 131518.
- [52] A. E. Clements, A. S. Hurn, and Z. Li, "Strategic bidding and rebidding in electricity markets," *Energy Econ.*, vol. 59, pp. 24–36, Sep. 2016.
- [53] T. Ji, Y. Yang, and Z. Jing, "Impacts of long-term transactions on strategic bidding in electricity market," *CSEE J. Power Energy Syst.*, vol. 9, no. 6, pp. 2090–2098, Nov. 2020.



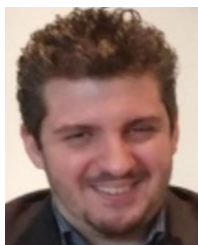
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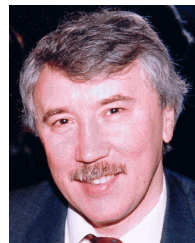
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